

Brain MRI Tumor Classification with ROI Segmentation

Progress Update Report

Course: ECEN 766: Algorithms in Structural Bioinformatics

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Abstract

Brain tumors are critical medical conditions that require early and accurate diagnosis. This project focuses on building an automated pipeline for brain MRI tumor classification using classical image processing and machine learning techniques. The system includes preprocessing, segmentation, feature extraction, and classification stages. A baseline pipeline extracts features from the full MRI image, while the proposed approach introduces a Region-of-Interest (ROI) segmentation step prior to feature extraction. The objective is to evaluate whether isolating tumor regions improves classification performance.

Availability:

[Github Repo Link](#)

1. Progress and Current Results (Focus)

1.1 Data Acquisition and Preprocessing:

MRI datasets were collected from publicly available sources and standardized for consistent processing. All images were resized to a fixed resolution and converted to grayscale where required. Noise reduction techniques such as median filtering were applied to remove artifacts, and intensity normalization was performed to improve contrast consistency across samples. Preprocessing outputs were visually inspected to ensure correctness and consistency across the dataset. In addition to basic preprocessing, attention was given to ensuring consistency across different MRI scans. Since MRI images may vary in brightness and contrast depending on acquisition conditions, normalization plays a critical role in reducing inter-sample variability. Median filtering was chosen over linear filters due to its ability to preserve edges while effectively removing noise, which is particularly important for maintaining tumor boundaries. Furthermore, preprocessing quality was validated by visually inspecting multiple samples to ensure that tumor regions were not distorted or suppressed. This step was essential to prevent errors from propagating into later stages such as segmentation and feature extraction, where inaccuracies can significantly impact classification performance.

1.2 Baseline Pipeline Implementation:

The baseline pipeline was implemented by extracting features from the entire MRI image. Feature extraction includes discrete wavelet transform (DWT) to capture multi-scale texture information and gray-level co-occurrence matrix (GLCM) features to represent spatial intensity relationships. Additional statistical features such as mean, variance, and entropy were also included. A Support Vector Machine (SVM) classifier was trained using these features. Initial evaluation indicates that the baseline pipeline is functional and provides a reasonable performance benchmark. The combination of discrete wavelet transform (DWT) and gray-level co-occurrence matrix (GLCM) features was chosen to capture complementary characteristics of the MRI images. While DWT provides multi-scale frequency decomposition that highlights texture variations across different resolutions, GLCM captures spatial relationships between pixel intensities, reflecting structural patterns in the image. These features are particularly relevant in medical imaging, where tumor regions often exhibit irregular texture and intensity distributions compared to surrounding healthy tissue. By combining both frequency-domain and spatial-domain features, the model gains a richer representation of the data, improving its ability to distinguish between tumor and non-tumor regions.

1.3 ROI Segmentation and Integration:

A lightweight ROI segmentation module was developed using thresholding techniques followed by morphological operations such as erosion, dilation, and hole filling. Connected component analysis was used to isolate the most likely tumor region. The segmented ROI was then used for feature extraction, replacing full-image feature computation. This pipeline was successfully integrated with the SVM classifier. One of the key motivations behind ROI segmentation is to reduce the influence of irrelevant background regions such as skull structures, cerebrospinal fluid, and healthy brain tissue. By focusing only on the tumor region, the feature extraction process becomes more targeted and less susceptible to noise introduced by non-relevant areas. However, ROI extraction is inherently challenging due to variability in tumor shape, size, and intensity across different MRI scans. This makes it necessary to balance segmentation accuracy with computational simplicity, especially when using threshold-based approaches. Improving segmentation robustness remains an important aspect of ongoing work.

1.4 Preliminary Results:

Initial experiments were conducted on a subset of the dataset to compare the baseline and ROI-based pipelines. The baseline model achieved an accuracy in the range of approximately 68–72%. The ROI-based pipeline achieved an improved accuracy in the range of approximately 72–76%. These results suggest that ROI-based feature extraction reduces the influence of irrelevant regions and improves classification consistency. Further evaluation using cross-validation is currently in progress to validate these observations. A qualitative analysis of the results indicates that the baseline model often misclassifies images where tumor regions are small or exhibit low contrast relative to surrounding tissue. In contrast, the ROI-based approach improves classification by isolating these regions, thereby enhancing the signal-to-noise ratio in the extracted features.

However, the observed improvement is moderate rather than significant, suggesting that segmentation quality directly influences classification performance. This highlights the importance of developing more robust segmentation techniques, which is being explored as part of the current extensions to the project.

2. Barriers and New Ideas

2.1 Barriers:

Several challenges were encountered during implementation. ROI segmentation performance is sensitive to intensity variations and threshold selection, leading to inconsistencies across different images. Noise and artifacts present in MRI scans can negatively affect segmentation quality and feature extraction. Combining multiple feature types increases dimensionality, which may introduce redundancy and impact classifier performance. Additionally, the SVM classifier requires careful parameter tuning to achieve optimal results. Another challenge observed was the variability in dataset quality, where some MRI images contained artifacts, intensity inconsistencies, or noise patterns that affected both segmentation and feature extraction stages. These variations can lead to unstable model performance across different subsets of the data. Addressing such inconsistencies is important for improving the generalization capability of the model.

2.2 New Ideas and Exploratory Work:

In addition to the proposed pipeline, exploratory efforts have been initiated to evaluate alternative segmentation techniques such as edge-based and region-based methods. Alternative classification approaches, including lightweight neural models, are also being explored to assess potential improvements. These extensions aim to provide a broader understanding of performance trade-offs and guide further optimization. Additionally, exploring multiple segmentation approaches allows for a comparative analysis of their effectiveness under different conditions. This helps identify which methods are more robust to variations in MRI data. Similarly, experimenting with alternative classification models provides insight into whether more complex models can improve performance without significantly increasing computational cost or reducing interpretability.

3. Action Items and Expected Outcomes

3.1 Action Items:

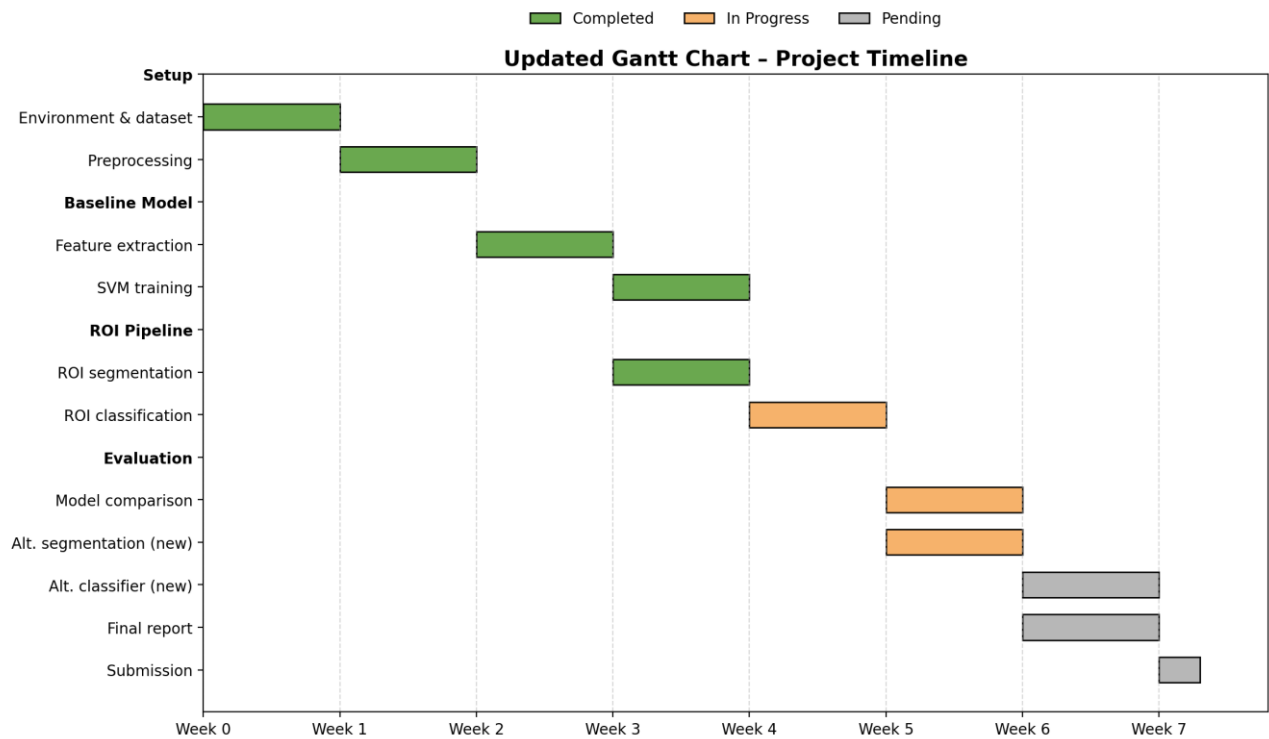
- Perform comprehensive evaluation using cross-validation
- Improve robustness of ROI segmentation through refined thresholding and morphological operations
- Compare baseline and ROI-based pipelines using consistent evaluation metrics
- Optimize feature selection and dimensionality reduction techniques
- Evaluate performance across different segmentation and classification methods

3.2 Expected Outcomes:

The final system is expected to demonstrate improved classification accuracy using ROI-based

segmentation. It will provide insights into the effectiveness of segmentation in medical image classification and highlight trade-offs between different approaches. The results will be summarized using metrics such as accuracy, precision, recall, and F1-score, along with visual representations such as confusion matrices.

4. Updated Gantt Chart



5. Individual Contributions

Tamil: Dataset acquisition and preprocessing, including image normalization and filtering

Yuvaraj: ROI segmentation implementation and validation across MRI samples

Karthigeyan: Feature extraction, classifier training, and initial evaluation